STARWEAVER®

Hands-On-Learning: Analyzing Phishing Attack with LLM

Scenario

You are an employee at a large corporation and have just received an email from the "IT Help Desk" with the subject line: "Urgent Security Update Required". The email states that a new security vulnerability has been discovered and that you must click a link immediately to update your system and avoid compromise. As a student of AI in cybersecurity, you decide to analyze this email's content using an AI tool before taking any action.

Instructions

1. Access a free and publicly available Large Language Model (LLM) of your choice (e.g., Gemini, ChatGPT, Claude).

2. Copy and paste the following email text into the LLM's chat interface:

“Subject: Urgent Security Update Required

Dear Employee,

This is an automated notification from the IT Help Desk. Our records indicate that your workstation has not been updated with the latest security patch, which addresses a critical vulnerability. This vulnerability could allow unauthorized access to your data and the company network.

To ensure the security of your account and the integrity of our systems, you must install the update immediately. Please click the link below to proceed with the update process.

http://security-update-it-support.com/patch

Failure to comply within 24 hours will result in the suspension of your network access to prevent further risk.

Thank you,

IT Help Desk”

3. Use the following prompt to instruct the AI: "Analyze the email above for signs of phishing and deceptive language. List the specific elements that are suspicious and explain why."

4. Review the AI's output. Note the specific tactics the LLM identifies, such as the sense of urgency, the suspicious URL, and the generic greeting.

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.